

Scheduling Mainframe's Student Operators and AED's Analysis

BMAN73701 Python Programming for Business Intelligence and Analytics

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1 Scheduling Mainframe's Student Operators

1.1 Problem Statement

$$H_{d,o} \geq 0 \quad (1)$$

$$o \in \left\{ \begin{array}{l} \text{E. Khan, Y. Chen, A. Taylor,} \\ \text{R. Zidane, R. Perez, C. Santos} \end{array} \right\} \quad (\text{Student Operators}) \quad (2)$$

$$Op_B = \{\text{E. Khan, Y. Chen, A. Taylor, R. Zidane}\} \quad (\text{Bachelor's Students}) \quad (3)$$

$$Op_M = \{\text{R. Perez, C. Santos}\} \quad (\text{Master's Students}) \quad (4)$$

$$d \in \{\text{Mon, Tue, Wed, Thu, Fri}\} \quad (\text{Days of Operation}) \quad (5)$$

$$WageRate_o \quad \text{given in Table 1.1} \quad (\text{Hourly Wage}) \quad (6)$$

$$Avail_{d,o} \quad \text{given in Table 1.2} \quad (\text{Maximum Available Hours}) \quad (7)$$

$$\min \text{ Total Labour Costs} = \sum_d \sum_o H_{d,o} \times WageRate_o \quad (8)$$

subject to:

$$\sum_d H_{d,o} \geq 8 \quad \forall o \in Op_B \quad (\text{Bachelor's Minimum Commitment}) \quad (9)$$

$$\sum_d H_{d,o} \geq 7 \quad \forall o \in Op_M \quad (\text{Master's Minimum Commitment}) \quad (10)$$

$$\sum_o H_{d,o} = 14 \quad \forall d \quad (\text{Mainframe Operating Hours per Day}) \quad (11)$$

$$0 \leq H_{d,o} \leq Avail_{d,o} \quad \forall d, o \quad (\text{Daily Availability}) \quad (12)$$

Table 1.1: Hourly Wage Rates ($WageRate_o$)

Operator	Wage Rate (£/hour)
E. Khan	25
Y. Chen	26
A. Taylor	24
R. Zidane	23
R. Perez	28
C. Santos	30

Table 1.2: Daily availability $Avail_{d,o}$ (hours)

Operator	Mon	Tue	Wed	Thu	Fri
E. Khan	6	0	6	0	6
Y. Chen	0	6	0	6	0
A. Taylor	4	8	4	0	4
R. Zidane	5	5	5	0	5
R. Perez	3	0	3	8	0
C. Santos	0	0	0	6	2

1.2 Mathematical Model Formulation

1.3 Data and Parameters

1.3.1 Data Inputs

1.3.2 Implementation

1.4 Model Results and Analysis

1.4.1 Model Results

1.4.2 Sensitivity Analysis

1.5 Conclusion and Recommendations

2 AED's Analysis

2.1 Problem Statement

2.2 Methodology

2.3 Robustness

2.4 Result

2.5 Discussion and Conclusion